

UNDO TABLESPACE

➤ How to add undo datafile with size 1GB?

STEP 1: First check the undo datafile with size.

```
SQL> SELECT FILE_ID, TABLESPACE_NAME, FILE_NAME, BYTES/1024/1024 "SIZE_MB", MAXBYTES/1024/1024/1024 "SIZE_GB" FROM DBA_DATA_FILES
2  WHERE TABLESPACE_NAME='UNDOTBS1';
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	SIZE_MB	SIZE_GB
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	340	31.9999847

STEP 2: Add the new datafile in existing undo tablespace with size 1GB.

```
SQL> ALTER TABLESPACE UNDOTBS1 ADD DATAFILE '/u01/app/oracle/oradata/ORCL/undotbs02.dbf' SIZE 1G AUTOEXTEND ON;
Tablespace altered.
```

STEP 3: Check the undo tablespace

```
SQL> SELECT FILE_ID, TABLESPACE_NAME, FILE_NAME, BYTES/1024/1024 "SIZE_MB", MAXBYTES/1024/1024/1024 "SIZE_GB" FROM DBA_DATA_FILES
2  WHERE TABLESPACE_NAME='UNDOTBS1';
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	SIZE_MB	SIZE_GB
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	340	31.9999847
5	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs02.dbf	1024	31.9999847

➤ How to check undo retention parameter at Database level?

This query is used to check the UNDO_RETENTION value.

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	900

➤ How to increase undo retention polity for 1Day?

STEP 1: Check the undo retention value.

```
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	900

STEP 2: Edit the undo retention value from 900 to 1 day.

```
SQL> ALTER SYSTEM SET undo_retention=86400 SCOPE=MEMORY;
System altered.
SQL> SHOW PARAMETER UNDO_RETENTION;
```

NAME	TYPE	VALUE
undo_retention	integer	86400

Important Notes

- UNDO_RETENTION is the minimum desired retention period. Oracle may overwrite undo data sooner if the undo tablespace runs out of space.
- To help retain undo for a full day, ensure the undo tablespace has sufficient free space.
- If using a dedicated undo tablespace and you want Oracle to preserve unexpired undo as much as possible, you can enable RETENTION GUARANTEE:

ALTER TABLESPACE UNDOTBS1 RETENTION GUARANTEE;

➤ Commands to check undo tablespace utilization.

Use this command to check the undo tablespace utilization.

```
SELECT
  df.tablespace_name,
  ROUND(df.total_mb,2) AS total_mb,
  ROUND(df.total_mb - fs.free_mb,2) AS used_mb,
  ROUND(fs.free_mb,2) AS free_mb,
  ROUND(((df.total_mb - fs.free_mb)/df.total_mb)*100,2) AS pct_used
FROM
  (SELECT tablespace_name,
    SUM(bytes)/1024/1024 total_mb
   FROM dba_data_files
   GROUP BY tablespace_name) df,
  (SELECT tablespace_name,
    SUM(bytes)/1024/1024 free_mb
   FROM dba_free_space
   GROUP BY tablespace_name) fs
WHERE df.tablespace_name = fs.tablespace_name
AND df.tablespace_name = (
  SELECT value
  FROM v$parameter
  WHERE name='undo_tablespace'
);
```

```
SQL> SELECT
  df.tablespace_name,
  ROUND(df.total_mb,2) AS total_mb,
  ROUND(df.total_mb - fs.free_mb,2) AS used_mb,
  ROUND(fs.free_mb,2) AS free_mb,
  2      3      4      5      6      ROUND(((df.total_mb - fs.free_mb)/df.total_mb)*100,2) AS pct_used
FROM
  (SELECT tablespace_name,
    SUM(bytes)/1024/1024 total_mb
   FROM dba_data_files
   GROUP BY tablespace_name) df,
  (SELECT tablespace_name,
    SUM(bytes)/1024/1024 free_mb
   FROM dba_free_space
   GROUP BY tablespace_name) fs
7      8      9      10     11     12     13     14     15
WHERE df.tablespace_name = fs.tablespace_name
16     17 AND df.tablespace_name = (
  SELECT value
  FROM v$parameter
  WHERE name='undo_tablespace'
); 18     19     20     21
```

TABLESPACE_NAME	TOTAL_MB	USED_MB	FREE_MB	PCT_USED
UNDOTBS1	1364	29.25	1334.75	2.14

➤ How to Drop undo datafile?

```
SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024 "SIZE_MB",MAXBYTES/1024/1024/1024 "SIZE_GB" FROM DBA_DATA_FILES
2 WHERE TABLESPACE_NAME='UNDOTBS1';
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	SIZE_MB	SIZE_GB
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	340	31.9999847
5	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs02.dbf	1024	31.9999847

```
SQL> ALTER TABLESPACE UNDOTBS1 DROP DATAFILE '/u01/app/oracle/oradata/ORCL/undotbs02.dbf';
ALTER TABLESPACE UNDOTBS1 DROP DATAFILE '/u01/app/oracle/oradata/ORCL/undotbs02.dbf'
*
ERROR at line 1:
ORA-03262: the file is non-empty
```

Oracle does not support directly dropping an active undo datafile. To remove an undo datafile, create a new undo tablespace, switch the database to it using ALTER SYSTEM SET UNDO_TABLESPACE, and then drop the old undo tablespace with DROP TABLESPACE ... INCLUDING CONTENTS AND DATAFILES.

STEP 1: Create a new undo tablespace.

```
SQL> CREATE UNDO TABLESPACE UNDOTBS2 DATAFILE '/u01/app/oracle/oradata/ORCL/UNDOTBS2_01.dbf' SIZE 340M;
Tablespace created.
```

```
SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024 "SIZE_MB",MAXBYTES/1024/1024/1024 "SIZE_GB" FROM DBA_DATA_FILES
2 WHERE TABLESPACE_NAME IN ('UNDOTBS1','UNDOTBS2');
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	SIZE_MB	SIZE_GB
5	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs02.dbf	1024	31.9999847
2	UNDOTBS2	/u01/app/oracle/oradata/ORCL/UNDOTBS2_01.dbf	340	0
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	340	31.9999847

STEP 2: Switch to the New Undo Tablespace.

```
SQL> ALTER SYSTEM SET UNDO_TABLESPACE=UNDOTBS2 SCOPE=BOTH;
System altered.
```

STEP 3: Verify the Change.

```
SQL> SHOW PARAMETER undo_tablespace;
```

NAME	TYPE	VALUE
undo_tablespace	string	UNDOTBS2

STEP 4: Drop the Old Undo Tablespace.

```
SQL> DROP TABLESPACE UNDOTBS1 INCLUDING CONTENTS AND DATAFILES;
Tablespace dropped.
```

STEP 5: Verify Undo Tablespaces

```
SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024 "SIZE_MB",MAXBYTES/1024/1024/1024 "SIZE_GB" FROM DBA_DATA_FILES
2 WHERE TABLESPACE_NAME IN ('UNDOTBS1','UNDOTBS2');
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	SIZE_MB	SIZE_GB
2	UNDOTBS2	/u01/app/oracle/oradata/ORCL/UNDOTBS2_01.dbf	340	0